

Alaa Faraj

+970594528583 | alaaemadf4@gmail.com | Ramallah, Palestine | [linkedin.com/in/ala-faraj-2b26a2209](https://www.linkedin.com/in/ala-faraj-2b26a2209) | github.com/alaax159

Arabic: Native **English:** B2 Upper Intermediate (EFSET Certified)

Computer Engineering student with hands-on experience building and evaluating AI systems — RAG pipelines, backend services and machine learning models — through an AI engineering internship and independent projects. Works across Python and TypeScript stacks.

SKILLS

Languages: Python, C, Java, JavaScript, TypeScript, SQL

AI / ML: Deep Learning, NLP, RAG & Vector Search, Transfer Learning, Prompt Engineering, Model Evaluation & Error Analysis, Side-by-Side Comparison, Data Annotation & Labeling, Grounding & Hallucination Checks, Data Preprocessing & Augmentation

Frameworks / Libraries: TensorFlow, Keras, HuggingFace Transformers, scikit-learn, FastAPI, NestJS

Databases / Tools: PostgreSQL (pgvector, Drizzle ORM), MySQL, Database Design, Docker, Redis & BullMQ, Azure AI Foundry, Git

EXPERIENCE

Artificial Intelligence Engineer — Internship

Jul 2026 – Sep 2026

ASAL Technologies

Rawabi, West Bank (Hybrid)

- Designed and **evaluated** a RAG pipeline over FDA DailyMed drug labels: pgvector search with SQL-level filtering that scopes evidence to each patient's own medications, versioned embedding profiles, and an audited Azure AI Foundry gateway.
- Built the **grounding guardrails**: every AI answer is citation-validated against retrieved source text and can never override the deterministic checker — the same ungrounded-claim and forced-inference failures quality review must catch.
- Implemented a deterministic medication-safety engine (DDInter drug-drug interactions, allergy checks) with immutable runs; added prescription scanning via Azure Document Intelligence OCR. Built the backend end to end: NestJS/TypeScript REST APIs, PostgreSQL (Drizzle ORM), Redis/BullMQ, Docker.

PROJECTS

Moeen — AI Medication Management Platform | [GitHub](#)

Jul 2026 – Sep 2026

- Mobile app and backend that track prescriptions and doses, check drug interactions and allergies, and answer medication questions with cited, label-grounded AI guidance (ASAL Technologies internship).
- Technologies:** NestJS, TypeScript, PostgreSQL + pgvector, Drizzle ORM, BullMQ, Azure AI Foundry, Azure Document Intelligence, Docker.

Wathiq — Retrieval-Augmented Fact Verification for Arabic Claims | [GitHub](#)

2026

- Arabic NLP** fact-verification pipeline: retrieves evidence for a claim from a document collection and classifies it as *Supported*, *Refuted* or *Not Enough Info*, trained on the AraFacts and ANS datasets.
- Compared model outputs **side-by-side across retrieval and verdict variants**, inspecting per-claim differences and not only aggregate scores: BM25 (test R@20 $\approx$ 0.98, MRR $\approx$ 0.83) vs. a dense AraBERT bi-encoder with in-batch contrastive negatives, and TF-IDF + logistic regression vs. an AraBERT/MARBERT cross-encoder.
- Technologies:** Python, PyTorch, HuggingFace Transformers, AraBERT/MARBERT, FAISS, rank_bm25, scikit-learn.

Time-of-Day Classification — Multimodal ML Evaluation Study | [GitHub](#)

Jan 2026

- Hand-labeled and quality-reviewed** the dataset, judging ambiguous cases against a consistent labeling standard.
- Designed a controlled **comparison of three approaches** to one task: a KNN color-feature baseline, an EfficientNetV2-L transfer-learning CNN (87.4% test accuracy), and a RoBERTa text classifier (76.4%).
- Evaluated each with accuracy, F1-score and confusion matrices on an imbalanced dataset, diagnosed per-class failure modes, and improved F1 by 12% over the baseline through partial fine-tuning, augmentation and early stopping.

EDUCATION

Birzeit University

Bachelor of Computer Engineering

Sep 2022 – Jun 2027 (expected)

Ramallah, Palestine

CERTIFICATIONS

Intro to Machine Learning with TensorFlow (Nanodegree) | Udacity – Palestine Launchpad

Mar 2026

AI Programming with Python & TensorFlow (Nanodegree) | Udacity

Jul 2025